MILITARY SPECIFICATION

CHARGE, PROPELLING FOR T-5123E1 TEST SHELL

1. SCOPE

1.1 This specification covers special requirements not included on the drawings, and provides the quality assurance provisions for the fabrication of parts, assembling, and the preparation for delivery for one type of propelling charge used with T-5123E1 Test Shell and designated as Charge, Propelling for T-5123E1 Test Shell.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposals, form a part of this specification to the extent specified herein:

SPECIFICATIONS

MILITARY

MIL-P-223 - Powder, Black
MIL-P-14769(MU) - Propellant M30 Type I (Master Lot) for Use in Charge XM72 Zone 3.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-286 - Propellants Solid Sampling, Examination and Testing
MIL-STD-1167 - Ammunition Data Card
MIL-STD-1168 - Lot Numbering of Ammunition
MIL-STD-1231 - Surveillance Inspection of Industrial Stocks of Propellant in Storage

DRAWINGS

ORDNANCE CORPS

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7548965 - Automatic Air Test for Loaded Containers Details
8849912 - Cap, Igniter Protector (Paper) Assembly
8864691 - Pad, Igniter
8864692 - End
8865306 - Contr, Metal, Ammo for Charge, Propelling 155MM: XM72 Pkg and Mkg
8885038 - Propelling Charge for T-5123E1 Test Shell (Hot Velocity)
8885039 - Propelling Charge for T-5123E1 Test Shell (Cold Pressure)
ET-8885038 - Equipment Tabulation
ET-8885039 - Equipment Tabulation

(Copies of specifications, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials. The contractor shall furnish objective evidence (see 6.2) that the material used to fabricate any part complies with the specifications listed on the applicable drawing.

3.2 Assembly. The charge shall comply with all requirements specified on Drawing (dwgs) 8885038 and 8885039, with all requirements specified herein, and with the requirements of applicable specifications to the extent specified on the drawing and herein.

3.3 Surveillance inspection of bulk propellant. A surveillance inspection of the bulk propellant used in the charge assembly shall be performed in accordance with 4.4.1. If the bulk propellant has less than 1.2 percent by weight of stabilizer, the bulk propellant lot shall be rejected.

3.4 Moisture content of black powder. The moisture content of the black powder of the igniter charges and flash reducer charge at the time of loading shall not exceed 0.7 percent when tested as specified in 4.4.2.

3.5 Propellant. Each charge of propellant shall be weighed and check weighed to a tolerance of $\pm$ 0.10 ounces (see 4.4.3).

3.6 Air pressure test. Each container shall be subjected to an air pressure test. The container shall maintain a pressure of 3 $\pm$ 0.1 pounds per square inch for a minimum of 15 seconds without indication of leakage. When machine for testing metal containers (see dwg 7548965) is used, the requirement for maintaining pressure shall be reduced to a minimum of 5 seconds. After completion of the air test the plug for the air test hole shall be seated with a torque of 5 to 7 foot pounds. The air pressure test shall be conducted in accordance with test procedure in 4.4.4.

3.7 Workmanship.

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3.7.1 Plastic and metal characteristics. All components shall be free from cracks, splits, cold flow, shrinkage, cold shuts, inclusions, porosity, or any similar characteristic.

3.7.2 Threads. Threads shall be full and undamaged for the entire minimum length or depth as required on the applicable drawing.

3.7.3 Burrs. Parts shall be free from burrs.

3.7.4 Foreign matter. Parts shall be free from chips, dirt, grease, rust, corrosion, or any embedded foreign material.

3.7.5 Cleaning. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.7.6 Stitching. Stitching shall not be missing where required. Stitching shall be free from any breaks or skipped areas. Tight stitching which breakds when the material is pulled to a flat surface or specified contour, shall be rejected.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 First article inspection. Not applicable.

4.3 Product acceptance inspection.

4.3.1 Lot formation. A lot shall consist of parts and assemblies produced by one manufacturer in one unchanged process, in accordance with the same drawing, same drawing revision, same specification and same specification revision. Drawing, specification and process changes not affecting safety, performance, storage, or interchangeability, as determined by the Government, shall not necessitate changing the lot interfix number. The criteria and procedures for the assignment of lot numbers shall be in accordance with MIL-STD-1168. Each lot of propellant shall require a new lot number.

4.3.1.1 Inspection lot. An inspection lot shall be the quantity of parts, subassemblies, or assemblies from the same lot, which are submitted to inspection for Government acceptance.

4.3.2 Product inspection examination. Critical characteristics shall require 100 percent inspection of all items in the inspection lot. A sample shall be selected at random from each inspection lot in accordance with the applicable major or minor characteristic table and inspected for all characteristics in the applicable classification. Any characteristic in the sample found to be defective shall reject the inspection lot. Sequence for the inspection of characteristics in each classification need not be followed as long as all characteristic inspections are performed.

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Major characteristic table I

Lot size	Sample size
1- 155	100% inspection
156- 200	155
201- 250	175
251- 300	190
301- 350	200
351- 400	210
401- 450	220
451- 500	225
501- 550	230
551- 600	235
601- 650	240
651- 700	243
701- 750	247
751- 800	250
801- 900	254
901-1000	258
1001-1100	261
1101-1200	264
1201-1300	266
1301-1400	268
1401-1500	270
1501-1600	272

Minor characteristic table II

Lot size	Sample size
1- 25	100% inspection
26- 30	25
31- 35	26
36- 40	27
41- 45	28
46- 50	29
51- 55	30
56- 60	31
61- 65	32
66- 70	33
71- 75	34
76- 85	35
86- 95	36
96- 110	37
111- 130	38
131- 156	39
157- 190	40
191- 240	41
241- 340	42
341- 520	43
521-1120	44
1121 and over	45

4.3.2.1 Classification of characteristics. Critical, major, and minor characteristics are defined in MIL-STD-105. Inspection shall be conducted as follows:

4.3.2.1.1 Pad, igniter (see dwg 8864691 covering a detail on dwgs 8885038 and 8885039).

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Material (see 3.1)	Visual
Minor:		
201.	Color	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.2 Cap, igniter protector (paper) assembly (see dwg 8849912, sheet 1, covering a detail of dwgs 8885038 and 8885039).

Categories	Characteristics	Method of inspection
Critical: None defined.		

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Major:
101. Materials (see 3.1).....Visual
102. Stitching (see 3.7.6).....Visual/Manual
103. Cap fits snugly on specified cylinder diameter.....Manual
104. No parts missing; properly assembled.....Visual

Minor:
201. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.3 End (see dwg 8864692 covering a detail of dwgs 8885038 and 8885039).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Material (see 3.1).....Visual

Minor:
201. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.4 Body (see dwg 8885041 covering a detail of dwgs 8885038 and 8885039).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Stitching (see 3.7.6).....Visual/Manual
102. Materials (see 3.1).....Visual

Minor:
201. Marking.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.5 Propelling charge for T-5123E1 test shell (Hot velocity) (see dwg 8885038) and Propelling charge for T-5123E1 test shell (cold pressure) (see dwg 8885039) before assembly of cap, igniter, protector assembly.

Categories	Characteristics	Method of inspection

Critical:
1. Weight of propellant (see 3.5).....Gage

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Major:
101. Weight of black powder.....Gage
102. Materials (see 3.1).....Visual
103. Diameter, max.....Gage
104. Length, max.....*Measure
105. Stitching (see 3.7.6).....Visual/Manual
106. Assembly not damaged or defective to extent powder charge is escaping or likely to escape during subsequent handling.....Visual
107. Igniter charge present.....Visual
108. Igniter color correct.....Visual

*Measure with suitable scale.

Minor:
201. Marking.....Visual
202. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.6 Propelling charge for T-5123E1 test shell (hot velocity) (see dwg 8885038) and Propelling charge for T-5123E1 test shell (cold pressure) (see dwg 8885039).

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Cap, igniter protector assembly properly assembled and fits snugly.....Visual/Manual

Minor:
201. Workmanship (see 3.7).....Visual/Manual

4.3.2.1.7 Container, metal, ammo for charge propellant XM72 (see dwg 8865306 and Section 5) prior to packing charge in container.

Categories	Characteristics	Method of inspection

Critical: None defined.

Major:
101. Wrapper present and snugly wound around propelling charge assembly.....Visual
102. Pad in bottom of container present.....Visual

Minor:
201. Workmanship (see 3.7).....Visual/Manual

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4.3.2.1.8 Container, metal, ammo for charge propellant XM72 Packing and marking (see dwg 8865306) prior to sealing cover.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Charges tightly packed	Manual
Minor:		
201.	Igniter protective cap near open end of container each charge	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.2.1.9 Container, metal, ammo for charge propellant: XM72 Packing and marking (see dwg 8865306) after sealing cover.

Categories	Characteristics	Method of inspection
Critical: None defined.		
Major:		
101.	Cover sealing bolt tight (bolt does not move in assembly direction when hand pressure is applied)	Visual/Manual
102.	Torque on test plug, min after air test (see 3.6)	Gage
103.	Car seal present and properly secured	Visual/Manual
104.	Test strip present	Visual
105.	Marking	Visual
106.	Charges tight in container	Manual
Minor:		
201.	Protective finish	Visual
202.	Workmanship (see 3.7)	Visual/Manual

4.3.3 Testing. The contractor shall perform all the tests specified in this paragraph. Tests do not have to be performed in the order specified.

Test	Test classification
a. Surveillance test	Critical
b. Moisture content of black powder	Major
c. Air pressure	Major
d. Moisture content of wrappers and pads	Major

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Of these tests a, b and d are considered destructive and samples so tested shall not be returned to the lot. The destructive samples are over and above the deliverable quantity of charges required by the contract.

4.3.3.1 Surveillance test (see 3.3). A sample or samples selected at random from the bulk lot of powder shall be tested in accordance with 4.4.1. If the samples fail to comply with the requirement 3.4, the bulk lot of powder shall be rejected and not used for Charge Assembly.

4.3.3.2 Moisture Content of Black Powder (see 3.4). One ounce sample minimum black powder selected at random from each lot used in eight hours production or fraction thereof shall be subjected to this test (see 4.4.2). If any sample fails to comply with 3.5 the production represented by the sample shall be rejected.

4.3.3.3 Air pressure test (see 3.6). Each container shall be subjected to this test (see 4.4.4). If any container fails to comply with 3.7, it shall be classified defective and removed from the lot.

4.3.3.4 Moisture Content of Wrappers and Pads (see dwg 8865306). A four ounce sample minimum of each paper product utilized in each eight hour production or fraction thereof shall be selected at random and subjected to this test (see 4.4.5). If any sample fails to comply with the requirements of dwg 8865306, the production represented by the sample shall be rejected.

4.3.4 Acceptance inspection equipment.

4.3.4.1 Critical and major characteristics and tests. Inspection and testing shall be performed with the acceptance inspection equipment, operating instructions, and calibration instructions specified on the Equipment Lists (EL). EL's for assemblies are listed on the Equipment Tabulation, ET-8885038 or ET-8885039 as applicable. When either of the conditions listed below exist, the contractor shall design required equipment or specify suitable commercial equipment. The contractor shall obtain approval of such designs, operating instructions, and special calibration instructions as specified in 6.3 prior to use on the contract.

a. Any deviation from equipment or procedures specified on the EL is desired.
b. The code "CD" is listed in the "Part or Identifying Number" column on the EL.

4.3.4.2 Minor characteristics and tests. Acceptance inspection equipment for minor characteristics is not specified on EL's. The acceptance inspection equipment, operating instructions and calibration instructions used by the contractor for minor classification inspection shall be approved by the Government representative authorized by the procuring agency.

4.4 Test methods. The tolerances specified herein are absolute with no allowance for test equipment inaccuracy. The tolerances used by the manufacturer shall be equal to the absolute tolerances less the accuracy tolerances of the test equipment used.

4.4.1 Surveillance test (see 3.3). Perform the surveillance test as specified in MIL-STD-1231. If the test conforms to requirements of MIL-STD-1231 accept the bulk powder and use in charge assemblies. If the test fails to comply with requirements select a sample at random from the bulk powder and test in accordance with MIL-STD-286 Method 201.4.

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4.4.2 Moisture content of black powder (see 3.4). The contractor shall provide adequate production controls to assure compliance of each lot with requirement 3.5. The moisture content shall be determined in accordance with specification MIL-P-223, paragraph 4.4.2.

4.4.3 Propellant (see 3.5). Each propellant charge shall be weighed and check-weighed. The check-weighing shall be accomplished independently of the original weighing using a different balance from that used to make the original weighing and, if performed manually, shall be performed by another operator.

4.4.4 Air pressure test (see 3.6). The test plug shall be removed from the container after propellant charges have been packed. The container shall be pressurized to 3.0 + 0.3 pounds per square inch gage as indicated by a pressure measuring device which has a resolution of 0.1 pounds per square inch. The air supply shall be shut off. An indicated pressure drop in excess of 0.1 pounds per square inch, on the pressure measuring device from the original reading when the air is shut off after a minimum 15 seconds time interval shall be cause for non compliance with requirements in 3.6. When the machine for air testing metal containers is used (see dwg 7548965), the minimum time interval shall be set at 5 seconds. After completion of the test the test plug shall be assembled to the container, after coating threads of plug and hole with graphite grease with a torque of 5 to 7 foot pounds.

4.4.5 Paper products (see 4.3.3.4). The contractor shall provide adequate production controls to assure the compliance of each lot with the specified requirements. The moisture content of the paper products shall be determined by obtaining a sample of approximately 60 grams of the material from the sample provided and accurately determine the exact weight of the sample to .0001 gram and record. Place sample in a moisture teller (Harry W. Dietest Co. of Detroit, Michigan or an approved equal) and dry for 30 + 1 minutes. After drying determine the weight of the dried sample to .0001 gram and record the weight determined. Again dry the sample in the moisture teller for 15 + 1 minute and determine the weight of the dried sample to .0001 grams. If the difference in the first dried weight and the second dried weight is 0.2 percent or less, the sample shall be considered dried, otherwise continue 15 + 1 minute drying and weighing until the 0.2 percent or less difference level is reached. The loss in weight shall be calculated as follows:

$$\text{Percent Moisture} = \frac{100\ (A-B)}{A}$$

Where A = weight of sample before drying
B = weight of sample after drying

The percent moisture of the sample shall comply with the requirements stated on dwg 8865306.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging, packing and marking. The charge shall be preserved, packaged, packed and marked in accordance with dwg 8865306 with the following exception:

1. Delete charges for Zones 1 and 2 and substitute charge Zone 3.

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6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

a. Title, number, and date of this specification.

6.2 Objective evidence. Records of contractor quality control and inspections which can be verified.

Custodian:
Army - MU

Preparing activity:
Army - MU

Project No. 1145-A-004

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